

Businesses can ask themselves the following questions to understand how ready they are for AI agent adoption.

- How does an AI agent align with their overall business goals and strategic objectives?
- Do they have a governance framework in place to manage the deployment and ethical use of an AI agent?
- How does the AI agent help in enhancing existing processes and enterprise strategy?
- What strategies does the enterprise have in place to identify, assess, and

mitigate risks associated with the AI agent?

- How will the enterprise manage the organisational changes required for the adoption of the AI agent?
- What key performance indicators (KPIs) will be used to measure the success and impact of AI agent initiatives?
- Is the current IT infrastructure capable of supporting the computational demands of the AI agent?
- Does the workforce possess the

skills to use an AI agent, and what are the implications for talent acquisition and upskilling?

- How will the enterprise address security and compliance concerns related to the deployment of an AI agent?
- Table 1 compares AI agents with traditional software.

The key benefits of AI agents are:

- Automate repetitive tasks to save time and manual work
- Analyse data and identify patterns quickly

Characteristic	AI agent	Traditional software
Behaviour	Proactive, learning and adapting	Reactive and does not learn from user behaviour
Business logic	Machine learning- and LLM-based; uses a reasoning-based approach to decide the steps needed to achieve a goal	Predefined rules and workflows; follows hardcoded, rule-based logic without deeper reasoning
Frameworks	Operates on frameworks like LangGraph, Auto-Gen, CrewAI, TensorFlow and PyTorch that support dynamic learning and adaptation; these frameworks are flexible and modular	Operates on frameworks and architectures like JADE, Cougaar, FIPA-OS; these frameworks are static and rigid
Communication	Dynamic function calling; leverages advanced APIs; utilises NLP to understand and generate human language, enabling smooth and intuitive interactions	Performs manual tasks; leverages standard APIs; rule-based systems for communication, which can be rigid and less intuitive
Autonomy	Highly autonomous; capable of reasoning and decision making	Limited autonomy; follows pre-programmed rules
Decision making process	Can plan and break down complex tasks into smaller steps for better solutions	Direct logic with no real reasoning capabilities
Scalability	High	Limited
Learning	Continuous learning and improvement	No learning capability or limited learning within a fixed set of responses
Tool access	Can access APIs and databases to enhance responses	Does not access external tools or systems
Problem solving	Handles complex, multi-step problems by combining reasoning with external resources	Handles simple, well-defined problems with scripted responses
Response generation	Iterates on responses by gathering more data and refining the solution until accurate	Provides an immediate response without revisiting or improving the result
Complex query handling	Capable of solving highly complex or ambiguous queries by using multiple resources	Best suited for straightforward, well-defined queries
Use cases	Customer service, predictive analytics, automation, multimodal interactions	Data entry, basic reporting, routine operations, single-mode interaction

Table 1: AI agents versus traditional software